

Coding Theory

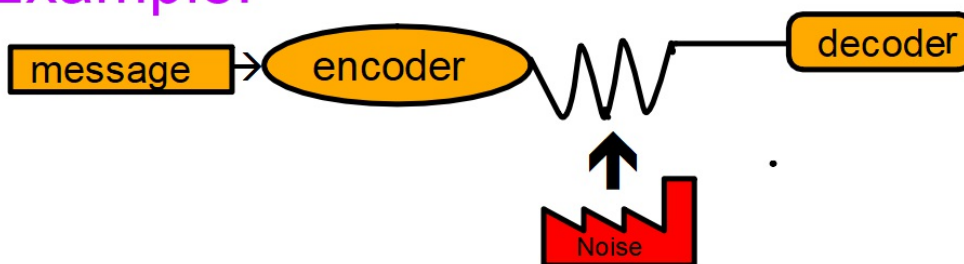
Coding theory deals with minimizing the noise of the conveyed message due to noise and to retrieve the original message.

Encodes and Decodes

Encode-transforms the incoming messages in such a way that the preserve of noise in the transform messages is detectable.

Decode-transforms the encoded their original form that can be understood by the receiver.

Example:



Commuication system

Group code

If $B = \{0, 1\}$ then $B^n = \{x_1, x_2, \dots, x_n / x_i \in B, i = 1, 2, 3, \dots, n\}$ is a group with $+_2$. This group $(B^n, +_2)$ is called a group code.

→ $(0, 0, 0, \dots, 0)$ is an identity of B^n .

→ The inverse of $x_1, x_2, \dots, x_n$ is itself.

Note: Any code which is a group with $+_2$ is called a group code.

Weight of a string

Let $x = x_1, x_2, \dots, x_n \in B^n$ be a string. Then the number of 1's in x is called the weight of x and it is denoted by $|x|$.

If $B=\{0,1\}$ then the possible

2-digit sequences: 00,01,10,11

3-digit sequences:

000,010,001,100,101,110,011,111

etc.

Hamming 1950-first complete error-detecting and error-correcting encoding procedure.

Let $x=(x_1,x_2,\dots,x_n)$ and $y=(y_1,y_2,\dots,y_n)$ be two n -tuples representing messages $x_1x_2\dots x_n$ and $y_1y_2\dots y_n$ respectively, where $x_i,y_i \in B$ for all i .

The **hamming distance between x_i and y_i** , denoted by $H(x,y)$, is the number of coordinates for which all x,y are different.

$\underline{\text{Ex}}$ $x = 1011 \Rightarrow$ $\{0,1\} \rightarrow$
 $y = 0101$ length as
 the word

$f_1(x, y) = |x +_2 y|$ $\begin{matrix} \text{no. of} \\ \text{ones} \\ \text{in} \\ x \end{matrix}$

$\begin{array}{r} 1011 \\ 0101 \\ \hline 1110 \end{array} +_2 = \begin{array}{r} 1011 \\ +_2 0101 \\ \hline 1110 \end{array}$

$f_1 \rightarrow \text{no. of ones} = 3$

·clearly, $H(x,y)=\sum(x_i+y_i)$.

Example: $x=1011$ and $y=0101$

Then $H(x,y)=3$,

minimum distance

The minimum distance of a code, whose words are n -tuples, is the minimum of the Hamming distance between all pairs code words in that code.

Example: find the minimum distance for the following code words, $x=(1,0,0,1)$, $y=(0,1,0,0)$ and $z=(1,0,0,0)$. Now, $H(x,y)=3$, $H(y,z)=2$ and $H(x,z)=1$. Therefore the minimum distance between these words is 1.

Group Code is an (m,n) encoding function $e:B^m \rightarrow B^n (n>m)$ which is 1-1. If $b \in B^m$ is the original word then $e(b)$ is the code word or encoded word representing b .

Matrix Notation:

An encoding function $e:B^m \rightarrow B^n (n>m)$ is given by a $m \times n$ matrix G over B . Then G is called a Generator matrix.

$$G = [I_m | A]$$

where A is $m \times (n-m)$ matrix to be chosen suitably.

If $w \in B^m$ then the code word is $e(w) = wG$,
where w is a vector

A **parity check matrix** is defined by $H=[A^T|I_{n-m}]$, which is $(n-m) \times n$ matrix.

Note:

1. H does not contain column of only 0's
2. No two columns of H are the same
3. H is unique.

Problem Find the code words generated by the encoding function $e: B^2 \rightarrow B^5$ w.r.t the parity check matrix H is

$$\begin{matrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

Rewrite the given parity check matrix in to $H=[A^T|I_{n-m}]$.
Here $m=2$ and $n=5$. Then the matrix H is

$$\begin{array}{cc|cc} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}$$

and hence the generator matrix $G=[I_m|A]$ is

$$\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array}$$

$$B^2 = \{00, 01, 10, 11\}$$

Let $e: B^2 \rightarrow B^5$ be an encoding function.

If $w \in B^m$ then $e(w) = wG$, where $w_{1 \times m}$ vector.

Therefore, the code words are

$$e(00) = 00000$$

$$e(01) = 01011$$

$$e(10) = 10011$$

$$e(11) = 11000 \text{ generated by } H.$$

$$\begin{array}{ccccc} 1 & 1 & \bullet & 1 & 0 & 0 & 1 & 1 \\ & & & 0 & 1 & 0 & 1 & 1 \end{array} = \begin{array}{cccccc} 1+0 & 0+1 & 0+0 & 1+1 & 1+1 & \\ & & & & & \end{array}$$

$$= 1 \ 1 \ 0 \ 0 \ 0$$

Problem Find the code words generated by the encoding function $e: B^3 \rightarrow B^6$ w.r.t the parity check matrix H is

$$\begin{matrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

Correcting a single error in transmission

Single error in a word say 0000 produce another word 0001,0010,0100,1000.

H provides a decoding scheme that correct a single error in transmission.

1. If r is received word ($1 \times n$) matrix and

$$H \bullet r^T = [0]$$

then there is no error.

2. If $H \bullet r^T = i$ th column of H

then there is a single error has occurred.

3. If neither case 1 nor case 2 occur then we conclude that more than one transmission error have occurred

1	0	1	0	0	$\begin{matrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{matrix}$	$1+0+1+0+0$
1	1	0	1	0	$\bullet$	$1=1+1+0+0+0$
0	1	0	0	1	$\begin{matrix} 0 \\ 1 \end{matrix}$	$0+1+0+0+1$

Examples

1. if $r=[1 \ 1 \ 1 \ 0 \ 1]$ then $H \bullet r^T = [0]$

$= 0 \ 0 \ 0$

2. if $r=[1 \ 1 \ 0 \ 1 \ 1]$ then $H \bullet r^T = [1 \ 1 \ 0]^T$
 (first column equal)

single error is 110

Correcting a single error is 010.

Examples

3. if $r = [1 \ 1 \ 0 \ 1 \ 0]$ then $H \bullet r^T = [1 \ 1 \ 1]^T$

a. $H \bullet r^T \neq$ any column of H

b. $[1 \ 1 \ 1]^T = [1 \ 1 \ 0]^T + [0 \ 0 \ 1]^T$
= column 1 of H + column 5 of H .

Therefore, 2 errors have occurred.

$$\begin{array}{ccccccccc} n=3 \ m=5 & 1 & 0 & 1 & 0 & 0 & \begin{array}{c} 1 \\ 1 \end{array} & 1+0+0+0+0 & 1 \\ & 1 & 1 & 0 & 1 & 0 & \bullet 0 & =1+1+0+1+0 & =1 \\ & 0 & 1 & 0 & 0 & 1 & \begin{array}{c} 1 \\ 0 \end{array} & 0+1+0+0+0 & 1 \end{array}$$

Problem

An encoding function $e: B^2 \rightarrow B^5$ is given by the generator matrix G

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

- Determine all code words.
- find the associated parity-check matrix H
- Use H to decode the received words 11101, 11011.

Solution: WKT $G = [I_m | A]$ and here $m=2$ and $n=5$. Therefore, the matrix I_2 and A are

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

a.

$B^2 = \{00, 01, 10, 11\}$.

Let $e: B^2 \rightarrow B^5$ be an encoding function.

If $w \in B^m$ then $e(w) = wG$, where $w_{1 \times m}$ vector.

Therefore, the code words are

$e(00) = 00000$

$e(01) = 01011$ $\begin{matrix} 1 & 0 & \bullet & 1 & 0 & 1 & 1 & 0 \\ & & & 0 & 1 & 0 & 1 & 1 \end{matrix} = 1+0 \quad 0+0 \quad 1+0 \quad 1+0 \quad 0+0$

$e(10) = 10010$

$e(11) = 11000$ generated by G . $= 1 \ 0 \ 0 \ 1 \ 0$

b.

WKT the parity check matrix is $H=[A^T|I_{n-m}]$.

Then the matrix H is

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

We observe that H contains no columns of 0's and no column of H are identical.

Therefore, H corrects single error in transmission.

c.

For the code word $r=11101$, then

$$H \bullet r^T = [0 \ 0 \ 0]^T$$

Since zero matrix, the decoded message is got by retaining the first two components of r . Therefore, decode message is $11(11101)$.

d. For the code word $r=11011$, then

$$H \bullet r^T = [1 \ 1 \ 0]^T = \text{first column of } H$$

Therefore, decode message is $01(01011)$.

$$\begin{array}{ccccc} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{array} \bullet \begin{array}{c} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{array} = \begin{array}{cc} 1+0+1+0+0=2 & 0 \\ 1+1+0+0+0=2 & 0 \\ 0+1+0+0+1=2 & 0 \end{array}$$